

Temperature Scales

Answer key with worked solutions

A. Understanding

1. The factor 9/5.

The same temperature interval is 180 °F or 100 °C, so 1 °C is the same size as $180/100 = 9/5 = 1.8$ **Fahrenheit degrees**. To convert, multiply the Celsius value by 9 °F/5 °C (and allow for the different zero points with the 32 °F).

2. Kelvin vs Celsius.

Any two of: kelvin has **no degree sign** (300 K, not 300 °K); kelvin **starts at absolute zero**, so it is never negative; kelvin must be used in gas-law and other calculations involving ratios of temperatures. (The size of one kelvin and one Celsius degree is the same.)

B. Celsius and Fahrenheit

3. To °F.

$$T_F = (9\text{ °F}/5\text{ °C}) \times T_C + 32\text{ °F}$$

$$(a) (9\text{ °F}/5\text{ °C})(100\text{ °C}) + 32\text{ °F} = 180\text{ °F} + 32\text{ °F} = \mathbf{212.0\text{ °F}}$$

$$(b) (9\text{ °F}/5\text{ °C})(-10\text{ °C}) + 32\text{ °F} = -18\text{ °F} + 32\text{ °F} = \mathbf{14.0\text{ °F}}$$

$$(c) (9\text{ °F}/5\text{ °C})(37.5\text{ °C}) + 32\text{ °F} = 67.5\text{ °F} + 32\text{ °F} = \mathbf{99.5\text{ °F}}$$

4. To °C.

$$T_C = (5\text{ °C}/9\text{ °F}) \times (T_F - 32\text{ °F})$$

$$(a) (5\text{ °C}/9\text{ °F})(50\text{ °F} - 32\text{ °F}) = (5\text{ °C}/9\text{ °F})(18\text{ °F}) = \mathbf{10.0\text{ °C}}$$

$$(b) (5\text{ °C}/9\text{ °F})(0\text{ °F} - 32\text{ °F}) = (5\text{ °C}/9\text{ °F})(-32\text{ °F}) = \mathbf{-17.8\text{ °C}}$$

$$(c) (5\text{ °C}/9\text{ °F})(451\text{ °F} - 32\text{ °F}) = (5\text{ °C}/9\text{ °F})(419\text{ °F}) = \mathbf{232.8\text{ °C}}$$

C. Kelvin

5. To kelvin.

$$(a) 25\text{ }^{\circ}\text{C} + 273.15 = \mathbf{298.15\text{ K}}$$

$$(b) -196\text{ }^{\circ}\text{C} + 273.15 = \mathbf{77.15\text{ K}}$$
 (about 77 K)

$$(c) \text{ First to } ^{\circ}\text{C}: (5\text{ }^{\circ}\text{C}/9\text{ }^{\circ}\text{F})(212\text{ }^{\circ}\text{F} - 32\text{ }^{\circ}\text{F}) = 100\text{ }^{\circ}\text{C}; \text{ then } 100\text{ }^{\circ}\text{C} + 273.15 = \mathbf{373.15\text{ K}}$$

6. 300 K.

$$T_{\text{C}} = 300\text{ K} - 273.15 = \mathbf{26.85\text{ }^{\circ}\text{C}}$$
 (26.8 °C to three significant figures: an exact 5 rounds to the even digit)

$$T_{\text{F}} = (9\text{ }^{\circ}\text{F}/5\text{ }^{\circ}\text{C})(26.85\text{ }^{\circ}\text{C}) + 32\text{ }^{\circ}\text{F} = 48.33\text{ }^{\circ}\text{F} + 32\text{ }^{\circ}\text{F} = \mathbf{80.3\text{ }^{\circ}\text{F}}$$

D. Temperature changes

7. A rise of 15 °C.

A kelvin is the same size as a Celsius degree, so the rise is **15 K**. In Fahrenheit, use only the size factor (the 32 °F offsets cancel for a change):

$$(1.8\text{ }^{\circ}\text{F}/^{\circ}\text{C})(15\text{ }^{\circ}\text{C}) = \mathbf{27\text{ }^{\circ}\text{F}}$$

E. Where the scales meet

8. -40.

Let the same number be x . Then $x = \frac{9}{5}x + 32$, so $x - \frac{9}{5}x = 32$, $-\frac{4}{5}x = 32$ and $x = \mathbf{-40}$.

$$\text{Check: } (9\text{ }^{\circ}\text{F}/5\text{ }^{\circ}\text{C})(-40\text{ }^{\circ}\text{C}) + 32\text{ }^{\circ}\text{F} = -72\text{ }^{\circ}\text{F} + 32\text{ }^{\circ}\text{F} = -40\text{ }^{\circ}\text{F} \checkmark$$

9. Kelvin and Fahrenheit.

Write °F in terms of K: $T_{\text{F}} = \frac{9}{5}(T_{\text{K}} - 273.15) + 32$. Set $T_{\text{F}} = T_{\text{K}} = x$:

$$x = \frac{9}{5}x - 491.67 + 32, \text{ so } -\frac{4}{5}x = -459.67 \text{ and } x = \mathbf{574.59}$$

So **574.59 K = 574.59 °F**, which is $574.59\text{ K} - 273.15 = \mathbf{301.44\text{ }^{\circ}\text{C}}$.

Kelvin and Celsius numbers are **never** equal: $T_{\text{K}} = T_{\text{C}} + 273.15$ always differ by 273.15, because the two scales have the same size of degree but different zero points.